

2024 Ph H2 Q4

Section: Our Dynamic Universe

Topic: Gravitation; Collisions & Impulse

Summary of Question

DART (mass 570 kg) collides with Dimorphos at 6.6 km s^{-1} ; Dimorphos orbits Didymos. We compare masses (orders of magnitude), calculate the gravitational force between Earth and Didymos when $r = 1.1 \times 10^{10} \text{ m}$, find the maximum kinetic energy that could be transferred in the collision, and sketch the force-time profile.

(a)(i) Orders of magnitude

Mass of Earth: $6.0 \times 10^{24} \text{ kg}$; Mass of Didymos: $5.3 \times 10^{11} \text{ kg}$.

Difference in powers of ten = $24 - 11 = 13$.

Final answer: Earth is 13 orders of magnitude more massive than Didymos.

(a)(ii) Gravitational force Earth-Didymos

Relationship: $F = G m_1 m_2 / r^2$.

Substitution: $F = (6.67 \times 10^{-11}) \times (6.0 \times 10^{24}) \times (5.3 \times 10^{11}) / (1.1 \times 10^{10})^2$

Calculation: $F \approx 1.75\text{e}+06 \text{ N} \approx 1.75 \times 10^6 \text{ N}$.

Final answer: $1.75 \times 10^6 \text{ N}$ (3 s.f.).

(b)(i) Maximum kinetic energy transfer

Assuming all spacecraft kinetic energy is transferred: $E_k = \frac{1}{2} m v^2$.

Substitution: $\frac{1}{2} \times 570 \times (6.6 \times 10^3)^2$.

Calculation: $E_k \approx 1.24\text{e}+10 \text{ J} \approx 1.24 \times 10^{10} \text{ J}$.

Final answer: $1.24 \times 10^{10} \text{ J}$.

(b)(ii) Force-time graph during collision

Sketch a short-duration, single-peaked pulse starting and ending at $F = 0$ (smooth rise to a maximum then fall).

Label axes $F \text{ (N)}$ and $t \text{ (s)}$. The area under the curve equals the impulse (Δp).

Final answer: Single-peaked pulse; area = impulse.

Revision Tips

Orders of magnitude: subtract exponents when comparing values in standard form.

When squaring r in gravitation, take care with powers of ten.

In impact questions, the area under the $F-t$ graph is the impulse.